

Reason about rounding boundaries

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: What is the smallest whole number that rounds to 86,000 to the nearest 1,000? Copy or complete the first step.

2. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

3. Is 2,650,000 rounded to the nearest million 3,000,000?

4. Miro says: Miro includes 85,499 in the range that rounds to 86,000. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: What is the smallest whole number that rounds to 86,000 to the nearest 1,000? Copy or complete the first step.

Answer 85,500.

2. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

Answer 4,349,999.

3. Is 2,650,000 rounded to the nearest million 3,000,000?

Answer Yes.

4. Miro says: Miro includes 85,499 in the range that rounds to 86,000. Is this correct? Explain.

Answer 85,499 is nearer 85,000. The lower boundary is 85,500.

Reason about rounding boundaries

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. What is the smallest whole number that rounds to 86,000 to the nearest 1,000?

6. Check this result using an inverse, estimate or known fact: Is 2,650,000 rounded to the nearest million 3,000,000?

2. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

7. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?

3. Is 2,650,000 rounded to the nearest million 3,000,000?

8. Identify and correct this misconception: Miro includes 85,499 in the range that rounds to 86,000.

4. Solve this independently and show every stage: What is the smallest whole number that rounds to 86,000 to the nearest 1,000?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

10. Write and solve a similar question to: Is 2,650,000 rounded to the nearest million 3,000,000?

Teacher answers and checking prompts

1. What is the smallest whole number that rounds to 86,000 to the nearest 1,000?
Answer 85,500.
2. A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?
Answer 4,349,999.
3. Is 2,650,000 rounded to the nearest million 3,000,000?
Answer Yes.
4. Solve this independently and show every stage: What is the smallest whole number that rounds to 86,000 to the nearest 1,000?
Answer 85,500.
5. Use a second method or representation for: A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?
Answer Expected result: 4,349,999. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Is 2,650,000 rounded to the nearest million 3,000,000?
Answer Expected result: Yes. A valid check must be shown.
7. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?
Answer 9,999, using 715,000 and 724,999.
8. Identify and correct this misconception: Miro includes 85,499 in the range that rounds to 86,000.
Answer 85,499 is nearer 85,000. The lower boundary is 85,500.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Is 2,650,000 rounded to the nearest million 3,000,000?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason about rounding boundaries

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?

6. Create a new example that tests the same idea as: Is 2,650,000 rounded to the nearest million 3,000,000?

2. Prove the answer to this question, rather than only calculating it: What is the smallest whole number that rounds to 86,000 to the nearest 1,000?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: Yes.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro includes 85,499 in the range that rounds to 86,000.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Two different numbers both round to 720,000 to the nearest 10,000. What is the greatest possible difference?
Answer 9,999, using 715,000 and 724,999.
2. Prove the answer to this question, rather than only calculating it: What is the smallest whole number that rounds to 86,000 to the nearest 1,000?
Answer Expected result: 85,500. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A number rounds to 4,300,000 to the nearest 100,000. What is the greatest possible number?
Answer Expected result: 4,349,999. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: Yes.
Answer One valid reconstruction is: Is 2,650,000 rounded to the nearest million 3,000,000? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro includes 85,499 in the range that rounds to 86,000.
Answer 85,499 is nearer 85,000. The lower boundary is 85,500.
6. Create a new example that tests the same idea as: Is 2,650,000 rounded to the nearest million 3,000,000?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.